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SPARC

Single-Pass Scaling for Motion Forecasting with Conformal Bayesian Last Layers

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One question: How can a fast motion forecaster know when its future should be trusted?

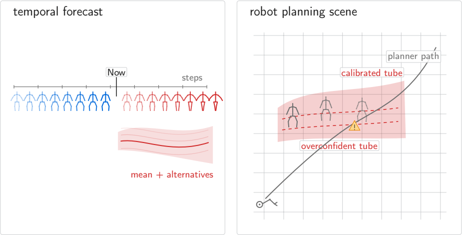


Fig. 1: A calibrated tube exposes risk that a point forecast hides.

The Missing Combination

- **Single-pass:** no MC sampling or ensemble loop at test time [2, 5].
- **Structured:** uncertainty must couple future time steps and body joints.
- **Calibrated:** nominal 95% tubes should cover about 95% under the stated assumptions [6, 7].

Gap: No existing recipe gives all three at once.

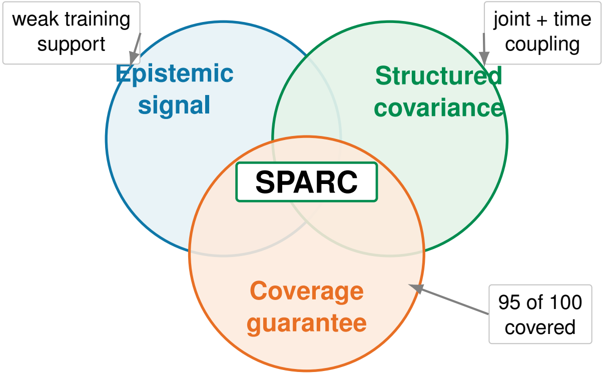


Fig. 2: SPARC targets epistemic sensitivity, structured uncertainty, and finite-sample calibration together.

SPARC in One Forward Pass

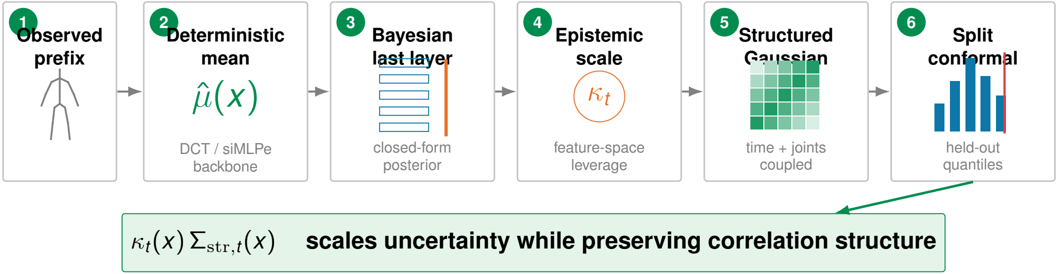


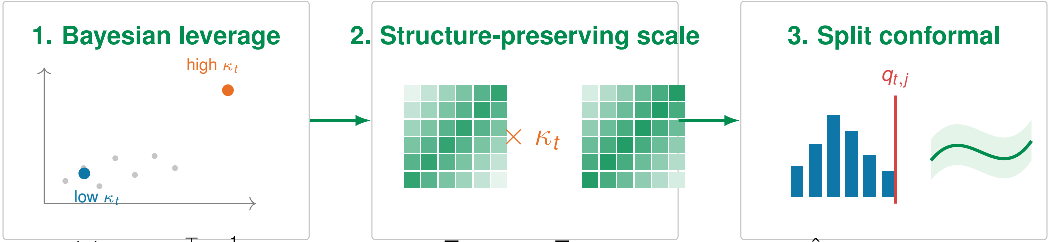
Fig. 3: A deterministic forecaster becomes a structured, calibrated density without stochastic test-time sampling.

The backbone supplies the mean; SPARC changes the uncertainty layer, not the forecast itself.

[1, 4]

WHY THE INTERFACE WORKS

From Feature Leverage to Valid Tubes



$\kappa_t(x) = 1 + \phi_t^\top \Lambda_{n,t}^{-1} \phi_t$ $\quad$ $\Sigma_{\mathrm{hyb},t} = \kappa_t \Sigma_{\mathrm{str},t}$ $\quad$ $\hat{C}_\alpha(x) = \hat{\mu} \pm q_{t,j}\hat{\sigma}$

Fig. 4: Analytic leverage scales a structured covariance; calibration converts the scale into a target-coverage tube.

Guarantee: 95% finite-sample **marginal** coverage under exchangeability [6, 7].

Not claimed: strong conditional, full-trajectory, or arbitrary-shift coverage [3].

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Evidence Across Nine Dataset/Protocol Blocks

1.00 NLL rank

2.69 MPJPE+NLL

2.50 W95(CP)

Model	NLL	MPJPE+NLL
MeanFT	6.11	4.17
Bridge-CQR	3.00	3.81
Deep ensemble	5.56	6.89
CoMusion	6.44	6.25
SPARD	6.56	6.83
SPARC	**1.00**	**2.69**

mean ranks; lower is better.

Table 1: Selected

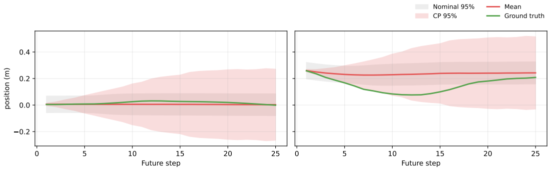

Fig. 5: Nominal and conformal 95% intervals around representative 3DPW trajectories.

Interpretation. Coverage matters only with informative width. SPARC retains efficient calibrated tubes in one forward pass.

SPARC is not a uniform point-error winner. Its consistent gain is **probabilistic**: better density quality and the best combined point+density rank.

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κ Turns Uncertainty Into a Risk Signal

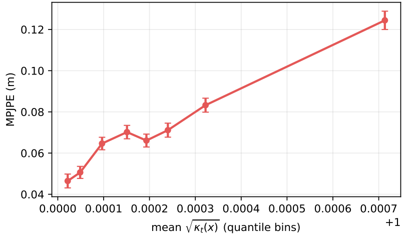

Fig. 6: Human3.6M MPJPE rises across bins of mean $\sqrt{\kappa_t(x)}$.

1.79×

MPJPE in the top-κ decile

- Filtering that decile reduces MPJPE by about 9%.
- The signal is available after a **single forward pass**.
- Thresholds are policy parameters tuned on held-out data, not a formal shift guarantee.

Takeaway: analytic κ_t + structured covariance + split CP = single-pass calibrated motion uncertainty. Full benchmark and ablations at our poster.

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Full Cross-Dataset Mean Ranks

Model	MR MPJPE	MR NLL	MR W95(CP)	MR MPJPE+NLL
siMLPe (base) [4]	7.89	10.67	–	9.28
MeanFT [4]	**2.22**	6.11	–	4.17
AuxTasks	9.06	11.78	–	10.42
Symplectic	16.22	15.78	–	16.00
DeFeeNet	7.61	10.44	–	9.03
GSPS	7.67	9.00	5.44	8.33
HumanMAC	10.22	10.00	–	10.11
SeSGCN	13.39	11.78	–	12.58
Deep ensemble [5]	8.22	5.56	5.39	6.89
Bridge-CQR	4.61	3.00	9.11	3.81
SkeletonDiffusion	15.00	13.67	9.78	14.33
BeLFusion	9.56	8.00	5.89	8.78
TransFusion	12.44	9.78	4.78	11.11
CoMusion	6.06	6.44	3.78	6.25
SPARD	7.11	6.56	3.39	6.83
MotionMap	16.11	16.22	9.44	16.17
SLD-HMP	13.22	15.22	6.50	14.22
SPARC	4.39	**1.00**	**2.50**	**2.69**

Table 2: Mean ranks across nine dataset/protocol blocks.

Strict Split Separation

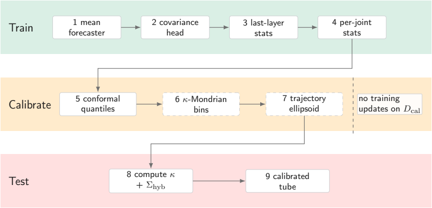

Fig. 7: Training fits parameters; calibration stores quantiles; evaluation is used once for metrics.

- No network, covariance, conjugate statistic, or hyperparameter is updated on the calibration or evaluation split.
- The exchangeability guarantee applies to the split-conformal marginal scores.

BACKUP · THE DEFAULT OPTIMIZES THE CALIBRATED RECIPE

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Component Ablation

Variant	κ_t	Hybrid	GraphJ	Coupled	MR MPJPE+NLL	MR W95(CP)
Base $\kappa(x)$	–	–	–	–	5.28	4.78
Timewise κ_t	✓	–	–	–	4.61	3.56
Hybrid MN	–	✓	–	–	3.94	4.00
Hybrid MN-GraphJ	–	✓	✓	–	**2.47**	4.78
Hybrid MN-GraphJ + κ_t	✓	✓	✓	–	<u>2.53</u>	3.78
SPARC coupled κ_t	✓	✓	✓	✓	3.17	**2.56**

Table 3: The coupled recipe gives the strongest calibrated tube-width rank within the ablation family.

Why this default?

Hybrid MN-GraphJ favors MPJPE+NLL, while the coupled SPARC recipe gives the best W95(CP) rank for the complete deployment objective.

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